Information, Not Architecture: Cross-Market News, Deep Learning and Tail-Risk Forecasting

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Abstract

Does deep learning improve forecasts of Value-at-Risk and Expected Shortfall when it is given more information, or only when it is given a better architecture? We study pre-open forecasts for 24 equity indices, using the returns of foreign markets that close before each market opens. All models are trained or evaluated with the strictly consistent FZ0 score. A pre-registered protocol develops the models on twelve indices and tests the frozen models exactly once on twelve further indices. Cross-market information is valuable: adding two overnight-news terms to a GJR-GARCH model lowers the loss at every risk level (Diebold–Mariano $t \approx -3.9$ in both samples), with the largest gains in Asian markets after U.S. and European sell-offs. Neural networks, however, do not extract this information. An attention-based contagion network, a neural news-impact GARCH with a proven stability property, and versions trained with an efficiency-improving composite score all fail to beat the two-parameter linear model. An ablation shows that the contagion network gains essentially nothing from the news, which the linear model turns into a large improvement. Forecast combinations that include both families are the most accurate in both samples and spend the fewest days in the Basel yellow zone. In tail-risk forecasting from daily data, information and inductive bias matter more than architecture.

Keywords: Expected Shortfall, Value-at-Risk, volatility spillovers, contagion, deep learning, elicitability, pre-registration.
JEL: C45, C52, C53, C58, G15, G17.

1 Introduction

Machine learning has changed empirical finance (Gu et al., 2020), and tail-risk forecasting is a natural candidate. Expected Shortfall (ES), the Basel market-risk measure, is jointly elicitable with Value-at-Risk (VaR) (Fissler and Ziegel, 2016). A neural network can therefore be trained directly on a strictly consistent loss (Patton et al., 2019). In a companion paper (Agrawal, 2026)

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we found that such networks match, but do not beat, the best econometric models when both see a market's own return history, and that they add information to forecast combinations. That leaves open a question with practical and economic content: is the bottleneck the *architecture*, or the *information*?

We answer it with a source of information that is available to every risk manager and largely ignored by univariate models: what other markets did overnight. Because markets trade in sequence across time zones, a risk manager in Tokyo or Taipei sets the morning VaR after the U.S. and European sessions have closed. Volatility spills over across markets (Engle et al., 1990; Hamao et al., 1990; Lin et al., 1994; Diebold and Yilmaz, 2009, 2012), and during crashes it may do so non-linearly (Forbes and Rigobon, 2002). Learning such non-linear, state-dependent transmission is exactly the kind of task deep learning is supposed to be good at.

We construct a pre-open information set that is free of look-ahead by design (Section 2). We then compare three ways of using it: a linear GJR-GARCH-X with skewed-t innovations, an attention-based contagion network, and a neural news-impact GARCH. To guard against the forking-paths problem that afflicts machine-learning studies, the design is pre-registered. Models are developed on twelve indices, frozen in a public, time-stamped commit, and then evaluated exactly once on twelve further indices that no model had seen.

Findings. First, cross-market information is valuable. Two news terms in a GARCH variance equation improve tail-risk forecasts at every level, with Diebold–Mariano statistics of about -3.9 in both samples. The gains are largest for Asian markets, consistent with overnight transmission from West to East. Second, the networks do not extract this information. None of the neural designs beats the linear model, and an ablation shows that the contagion network gains essentially nothing from the news that the linear model turns into a large improvement. Third, forecast combinations that include both families are the most accurate, and the companion paper's combination result replicates on the new markets ($t = -7.1$).

Contributions. (i) A look-ahead-free, pre-open information set for daily tail-risk forecasting across time zones. (ii) Two theoretical results for neural tail-risk models: a composite FZ0+QLIKE score that is strictly consistent and more efficient for learning dynamics (Proposition 1), and a neural news-impact GARCH with provable stability and exact nesting of GJR-GARCH (Proposition 2). (iii) Out-of-sample evidence, under a pre-registered protocol, that in this setting information and inductive bias matter more than architecture, together with the mechanism behind this: tail-focused losses learn dynamics inefficiently. (iv) Evidence on the direction of international tail-risk transmission.

2 Information set: what is known before a market opens

Equity markets in different time zones trade in sequence. When the risk manager of market i sets the Value-at-Risk for trading day D shortly before i opens, markets that have traded in the

meantime have already revealed information that i's own close-to-close history does not contain (Hamao et al., 1990; Lin et al., 1994). Let $C_i(D)$ and $O_i(D)$ denote the close and open of market i on its local trading day D, in coordinated universal time, and let D^- be i's previous trading day.

Definition 1 (Pre-open information set)**.** $\mathcal{F}^{\text{pre}}_{i,D}$ *contains the returns of market i up to D^- and, for every other market j, all returns of sessions s with $C_j(s) \le O_i(D) - 2m$, where m is a safety margin.*

The new information for day D is the overnight move of each foreign market,

$$x_{j,D} = \sum_{s:\, C_i(D^-) < C_j(s) \le O_i(D) - 2m} \frac{r_{j,s}}{s_{j,s}}, \tag{1}$$

where $s_{j,s}$ is j's trailing 250-day root mean square return up to the previous session, so that $x_{j,D}$ is measured in units of j's usual volatility. An indicator $a_{j,D}$ records whether at least one such session exists.

Historical exchange schedules have changed (Tokyo extended its close by 30 minutes in 2024, Seoul in 2016, Brazil abolished daylight saving in 2019). To rule out look-ahead bias, we use the *latest* plausible close of every exchange, including closing auctions, and the *earliest* plausible open, including pre-opening sessions. We then require an additional margin of $m = 30$ minutes on each side, and handle daylight saving with the IANA time-zone database. A violation could only arise from a schedule error of more than an hour in the conservative direction. The construction makes stale information look new, which can only weaken the predictive content of the news, never strengthen it. In our development sample, Asian markets see the previous U.S. and European sessions on about 97% of days, and U.S. markets see the same day's Asian sessions on about 95% of days.

3 Scoring and statistical efficiency

As in Agrawal (2026), forecasts of $(\text{VaR}_\alpha, \text{ES}_\alpha)$ for $\alpha \in \{1\%, 2.5\%, 5\%\}$ are trained and evaluated with the strictly consistent FZ0 loss of Patton et al. (2019),

$$L_\alpha(y, v, e) = -\frac{1}{\alpha e}\mathbf{1}\{y \le v\}(v - y) + \frac{v}{e} + \log(-e) - 1.$$

FZ0 receives information about its target mainly from tail days. Away from the tail, the loss is $v/e + \log(-e) - 1$, which depends on the forecast only through its level. Learning *dynamics*, i.e. how risk responds to news, from FZ0 alone is therefore statistically inefficient. A Gaussian quasi-likelihood instead learns the dynamics of volatility from every day's squared return. This motivates an auxiliary score.

Proposition 1 (Composite score)**.** *Let* $Q(y, \varsigma^2) = y^2/\varsigma^2 + \log \varsigma^2$ *(QLIKE). For* $\lambda \geq 0$*, the score*

$$S(y; \{v_k, e_k\}_{k=1}^K, \varsigma^2) = \sum_{k=1}^K L_{\alpha_k}(y, v_k, e_k) + \lambda\, Q(y, \varsigma^2)$$

is strictly consistent for the vector $(\{\mathrm{VaR}_{\alpha_k}, \mathrm{ES}_{\alpha_k}\}_{k=1}^K,\ \mathbb{E}[y^2])$*, provided* $\mathbb{E}y^2 < \infty$ *and every* L_{α_k} *is strictly consistent for its pair (Proposition 1 of Agrawal, 2026).*

Proof. $\mathbb{E}Q(Y, \varsigma^2) = \mathbb{E}[Y^2]/\varsigma^2 + \log \varsigma^2$ has derivative $(\varsigma^2 - \mathbb{E}Y^2)/\varsigma^4$ and is uniquely minimised at $\varsigma^2 = \mathbb{E}Y^2$. The expected composite score is a sum of terms that depend on disjoint components of the report, each uniquely minimised at the corresponding true functional, so the sum is uniquely minimised at the true vector. □

Remark 1. In a location-scale network $v_k = \varsigma q_k$, $e_k = \varsigma c_k$, the reports are linked. The composite score then still identifies the truth when the model is correctly specified. Under misspecification, λ trades off tail fit against volatility fit. We fix $\lambda = 1$ a priori and do not tune it. QLIKE is also the loss that is robust to noise in the volatility proxy (Patton, 2011).

4 Models

All models produce one-step-ahead $(\mathrm{VaR}, \mathrm{ES})$ forecasts at the three levels. The benchmarks of Agrawal (2026) (historical simulation, GARCH-type models with normal, Student-t and skewed-t innovations, filtered historical simulation, GARCH-EVT, CAViaR-ES and two FZ-GAS models) and its deep models (LSTM-FZ, TRF-FZ, GJR-LSTM) use only the close-to-close information set and are carried over unchanged. Four models are new.

GJRX-skt. An econometric model with the pre-open information: a GJR-GARCH(1,1,1) with Hansen skewed-t innovations whose variance responds linearly to squared foreign news,

$$\sigma_t^2 = \omega + (a + g\,\mathbf{1}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + b\,\sigma_{t-1}^2 + d_1 \bar{X}_t + d_2 \bar{X}_t^-, \tag{2}$$

where $\bar{X}_t$ is the average of $x_{j,t}^2$ over markets with fresh news and $\bar{X}_t^-$ the same average restricted to negative moves. Both are known before r_t is realised. The model is estimated by maximum likelihood with $d_1, d_2 \geq 0$ on the same rolling window and refit schedule as the other econometric models. It nests GJR-skt ($d_1 = d_2 = 0$).

NG-FZ (neural news-impact GARCH). Returns are scaled by a per-asset constant, $y_t = r_t/s$, and a d-dimensional state follows a *linear* recursion driven by a *learned* news-impact function

(Engle and Ng, 1993):

$$h_t = A\,h_{t-1} + N_\theta(y_{t-1}), \qquad A = \mathrm{diag}(a_1, \dots, a_d), \quad 0 < a_k < \rho < 1, \tag{3}$$
$$N_\theta(y) = B\,(y,\, y^2,\, y^2\mathbf{1}\{y<0\},\, |y|)' + c + g \cdot \mathrm{MLP}(\tanh(y/2), \log(1+y^2), \mathbf{1}\{y<0\}).$$

The first coordinate is a GJR-GARCH variance, the others are volatility memories at decay rates $0.5, \dots, 0.995$ (as in component GARCH). Log-variance and tail shape are read out by small networks, and the output uses the monotone head of Agrawal (2026).

Proposition 2 (Stability and nesting)**.** *(i) For any input path and any two initial states, $\|h_t - h'_t\|_\infty \le \rho^t \|h_0 - h'_0\|_\infty$: the filter forgets its initialisation exponentially (invertibility). (ii) If (y_t) is strictly stationary and ergodic with $\mathbb{E}\log^+\|N_\theta(y_0)\| < \infty$, then $h^*_t = \sum_{j\ge 0} A^j N_\theta(y_{t-1-j})$ converges almost surely, is the unique stationary ergodic solution of* (3)*, and $\|h_t - h^*_t\| \to 0$ exponentially almost surely (Brandt, 1986; Bougerol, 1993). (iii) With $g = g' = 0$ (both neural gates closed) and $d = 1$, NG-FZ is exactly GJR-GARCH(1,1) with a constant standardised tail.*

Proof. (i) $h_t - h'_t = A^t(h_0 - h'_0)$ because the input enters additively and does not depend on the state; $\|A^t\|_\infty = \max_k a_k^t \le \rho^t$. (ii) By the root test,

$$\limsup_{j\to\infty} \|A^j N_\theta(y_{t-1-j})\|^{1/j} \le \rho \cdot \limsup_{j\to\infty} \exp\Big(\tfrac{1}{j}\log^+\|N_\theta(y_{t-1-j})\|\Big) = \rho < 1$$

almost surely, because $\mathbb{E}\log^+\|N_\theta(y_0)\| < \infty$ implies $\log^+\|N_\theta(y_{-j})\|/j \to 0$ (Borel–Cantelli). Uniqueness and the convergence of h_t follow from (i). (iii) By substitution. □

The network is initialised at (iii), with the GARCH part at a fitted GJR model and the tail at the empirical tail of its residuals, so that training starts from a strong econometric model and learns corrections by minimising FZ0. Gradient descent on a linear recursion with bounded decays cannot produce explosive filters, which is a practical advantage over generic recurrent networks.

CTG-FZ (contagion network). CTG-FZ adds a cross-market module to LSTM-FZ, with identical inputs and hyperparameters otherwise. Each foreign market contributes a token $k_j = \mathrm{MLP}(x_j, |x_j|, \mathrm{emb}_j)$. A query is formed from the target market's embedding and its LSTM state, and attention over the markets with fresh news yields a contagion vector $c = \sum_j w_j k_j$, which is zero when no foreign market traded overnight. Because the tokens are non-linear in x_j and the query depends on the target's own volatility state, the module can represent disproportionate responses to large foreign crashes, market-specific lead–lag structure and state-dependent contagion. The weights w_j form an interpretable transmission matrix. In the final specification (v2, Section 5) the output is location–scale, $\mathrm{VaR} = s\,\varsigma\,q$ and $\mathrm{ES} = s\,\varsigma\,c$, with ς from a variance head trained by the composite score of Proposition 1.

LSTMQ-FZ (ablation). LSTMQ-FZ is identical to CTG-FZ v2 with the foreign news switched off. The comparison CTG-FZ versus LSTMQ-FZ therefore isolates the value of the cross-market

information for the network, holding architecture and objective fixed.

Combinations. COMB-All (all econometric models except HS, plus the three deep models of Agrawal, 2026) is extended to COMB-All+X by adding GJRX-skt and CTG-FZ, and to COMB-All+NG by adding NG-FZ. All combinations are equal-weight averages, specified before any result was seen.

5 Pre-registered design

Samples. The *development* sample consists of the twelve indices of Agrawal (2026): S&P 500, Nasdaq Composite, Russell 2000, FTSE 100, DAX, CAC 40, Nikkei 225, Hang Seng, BSE Sensex, S&P/ASX 200, KOSPI and Bovespa. The *holdout* sample consists of twelve further indices: S&P/TSX, SMI, TAIEX, Shanghai Composite, IPC, AEX, IBEX 35, KLCI, Jakarta Composite, Straits Times, ATX and BEL 20. The holdout markets were chosen by a mechanical rule, before any model was run on them. The rule requires at least 2,000 daily observations before 2006, no multi-month gaps and fewer than fifteen stale prices; it excluded OMXS30, Merval and FTSE MIB. Both samples use the out-of-sample period January 2006 to September 2026, a 2,000-day rolling window and the evaluation methods of Agrawal (2026).

Protocol. The protocol, its amendments and every development result are recorded in a public, time-stamped file (`docs/PROTOCOL.md` in the repository). New models were designed and revised on the development sample only. Before any holdout run, the final specifications were frozen in a git commit (`7a58b26`, 15 October 2026), and the holdout sample was then evaluated exactly once. The pre-registered hypotheses, at $\alpha = 2.5\%$ with panel Diebold–Mariano tests, were:

- H1 CTG-FZ has lower FZ0 loss than GJRX-skt (learning contagion versus a linear spillover model with the same information);
- H2 CTG-FZ has lower loss than LSTM-FZ (value of the cross-market information for the network);
- H3 COMB-All+X has lower loss than COMB-All.

Development history. For transparency we report every variant tried on the development sample (Table 1). NG-FZ v1 shared one GARCH block across markets. v2 gave each market its own block and overfitted. The diagnosis, that FZ0 learns volatility dynamics inefficiently, motivated v3 (v2 plus the auxiliary QLIKE score of Proposition 1), which roughly halved v2's deficit. Under the pre-specified rule (lowest development loss), v1 was frozen. CTG-FZ v1 reacted to foreign news but did not improve on LSTM-FZ. By the same efficiency diagnosis, v2 adopted the location–scale output with the composite score, and the ablation LSTMQ-FZ was added at that point. A bug in the first v2 run (a variable that silently disabled the new settings) was detected before any result was seen, fixed with a test, and the run was repeated. v2 was frozen by the selection rule.

Table 1: Development history (development sample, $\alpha = 2.5\%$): FZ0 loss relative to the stated comparison

Variant	Comparison	Loss difference	DM t
NG-FZ v1 (pooled GARCH block)	GJR-skt	+0.0044	0.42
NG-FZ v2 (market-specific block)	GJR-skt	+0.0272	2.18
NG-FZ v3 (v2 + QLIKE)	GJR-skt	+0.0146	1.65
CTG-FZ v1	GJRX-skt / LSTM-FZ	+0.0177 / +0.0016	1.30 / 0.96
CTG-FZ v2	GJRX-skt / LSTM-FZ	+0.0153 / −0.0007	1.22 / −0.29
CTG-FZ v2	LSTMQ-FZ (ablation)	+0.0002	0.48
GJRX-skt	GJR-skt	−0.0193	−3.90

Frozen for the holdout: NG-FZ v1, CTG-FZ v2, LSTMQ-FZ, GJRX-skt and all models and combinations of Agrawal (2026).

6 Results

The holdout sample was evaluated once, with the frozen models. We report the holdout results first; the development results (Table 1 and Appendix) tell the same story. Each sample contains about 62,000 asset-days per model and risk level.

6.1 Pre-registered hypotheses

Table 2 reports the three pre-registered tests and two auxiliary comparisons.

Cross-market information is valuable. GJRX-skt, which adds two news terms to GJR-GARCH, lowers the FZ0 loss relative to GJR-skt at every risk level. At 2.5% the panel Diebold–Mariano statistic is −3.9, the same as in the development sample, and the model is never significantly worse in any market. It is the best single model on the holdout and the only model with an MCS p-value of one at all three levels (Table 4).

The networks do not extract it (H1 and H2 not supported). CTG-FZ is not more accurate than GJRX-skt (H1: +0.012, $t = 1.3$). It is also not significantly more accurate than LSTM-FZ, which sees no foreign information at all (H2: −0.003, $t = -1.4$). The ablation is sharper still. CTG-FZ and its twin without the news, LSTMQ-FZ, differ by 0.0003 in average loss. Yet the information that the network leaves unused is worth 0.023 to a linear model. The same pattern appeared in the development sample (Table 1), so it is not an artefact of one set of markets.

Combinations dominate (H3 supported). Adding GJRX-skt and CTG-FZ to the paper-1 combination lowers the loss at every level (t between −4.8 and −6.3), in 10 to 11 of 12 markets and never significantly the other way. The paper-1 finding that the deep models add information to an econometric combination also replicates on the new markets: COMB-All versus COMB-Econ gives $t = -7.1$ at 2.5%, with 11 of 12 markets better.

6.2 Where the information matters

The gain from overnight news is concentrated where the timing argument predicts it should be. In the holdout, the largest improvements of GJRX-skt over GJR-skt occur in Jakarta, Singapore, Taiwan, Kuala Lumpur and Shanghai: Asian markets that open after the U.S. and European sessions

Table 2: Holdout sample: Pre-registered hypothesis tests (FZ0 loss of first model minus second)

Test (level)	Loss diff	Panel DM t	p-value	# assets better	# assets worse
H1: CTG-FZ vs GJRX-skt (1%)	0.0273	1.67	0.095	0	2
H1: CTG-FZ vs GJRX-skt (2.5%)	0.0125	1.27	0.206	0	0
H1: CTG-FZ vs GJRX-skt (5%)	0.0069	0.93	0.354	1	1
H2: CTG-FZ vs LSTM-FZ (1%)	-0.0039	-1.41	0.158	1	1
H2: CTG-FZ vs LSTM-FZ (2.5%)	-0.0026	-1.36	0.175	1	0
H2: CTG-FZ vs LSTM-FZ (5%)	-0.0032	-1.75	0.080	0	0
H3: COMB-All+X vs COMB-All (1%)	-0.0082	-4.80	0.000	10	0
H3: COMB-All+X vs COMB-All (2.5%)	-0.0054	-5.67	0.000	11	0
H3: COMB-All+X vs COMB-All (5%)	-0.0042	-6.27	0.000	11	0
Ablation: CTG-FZ vs LSTMQ-FZ (1%)	-0.0004	-1.10	0.273	0	0
Ablation: CTG-FZ vs LSTMQ-FZ (2.5%)	-0.0003	-1.06	0.290	0	0
Ablation: CTG-FZ vs LSTMQ-FZ (5%)	-0.0003	-1.38	0.168	0	0
Info: GJRX-skt vs GJR-skt (1%)	-0.0375	-3.51	0.000	4	0
Info: GJRX-skt vs GJR-skt (2.5%)	-0.0231	-3.88	0.000	5	0
Info: GJRX-skt vs GJR-skt (5%)	-0.0161	-4.00	0.000	6	0

Negative loss difference = the first model is more accurate. Panel Diebold–Mariano test on the cross-asset average loss differential.

have closed. The smallest occur in Europe and the Americas. The development sample shows the same ordering (Tokyo, Sydney and Hong Kong first, the DAX and CAC last). The estimated news coefficients mirror this. Negative foreign news raises next-day Asian variance strongly, while the coefficients for U.S. and European targets are close to zero. Tail risk is transmitted mainly from the West to Asia overnight, in line with the volatility-spillover evidence of Hamao et al. (1990), Lin et al. (1994) and Diebold and Yilmaz (2009). Here it is shown for Expected Shortfall and out of sample.

Figure 1 illustrates the mechanism for the Taiwan index. In March 2020, GJRX-skt raised the 2.5% VaR on the mornings after the large U.S. declines, *before* Taiwan's own losses. The univariate GJR-skt reacted only after them. CTG-FZ also responds to the overnight news, but erratically: it overshoots in some episodes and lags in others.

6.3 Accuracy, model confidence sets and calibration

Table 3 reports average losses for all models on the holdout. Three results of Agrawal (2026) carry over to the new markets. HS, GARCH with Gaussian or symmetric-t innovations, and the FZ-GAS models are significantly worse than GJR-skt. The pooled LSTM is at least as accurate as GJR-skt. The combinations that include the deep models dominate. The GJR-LSTM hybrid is significantly more accurate than GJR-skt on the holdout ($t = -2.8$), which was not the case in the development sample. NG-FZ is statistically indistinguishable from GJR-skt in both samples.

In the MCS (Table 4), GJRX-skt has a pooled p-value of one at every level. The recurrent neural models (LSTM-FZ, GJR-LSTM, NG-FZ, LSTMQ-FZ and CTG-FZ) and COMB-All+X are also retained in the pooled 90% set at the 2.5% and 5% levels. Every econometric model without the overnight information, the Transformer and COMB-All are excluded at every level. On the holdout markets, the recurrent networks are thus statistically closer to the best model than any

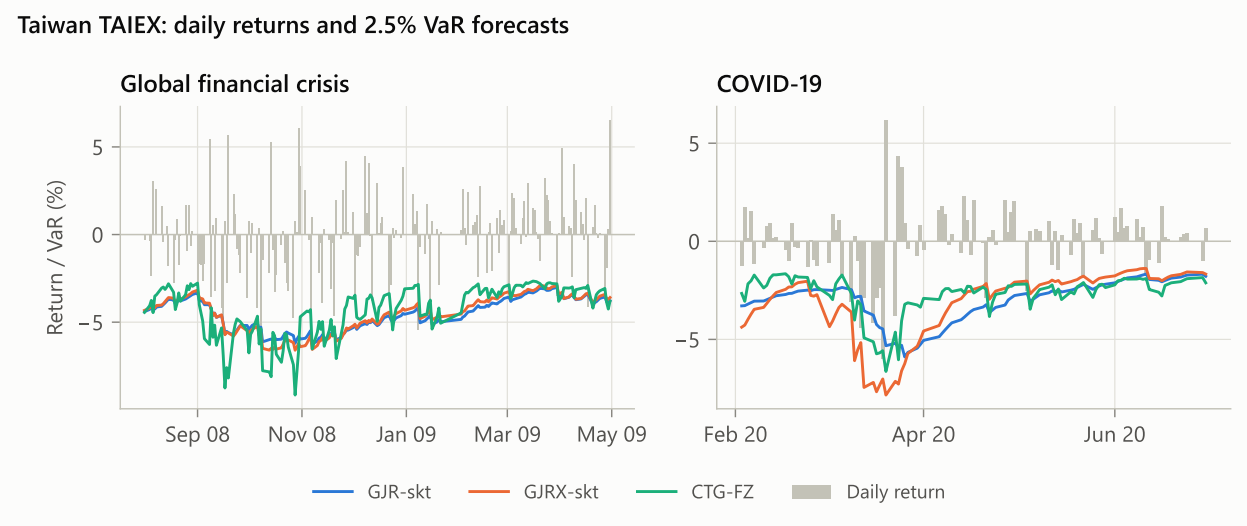


Figure 1: Holdout sample: Taiwan TAIEX daily returns and 2.5% VaR forecasts of GJR-skt (own history only), GJRX-skt and CTG-FZ (both with pre-open foreign news).

Table 3: Holdout sample: Average FZ0 loss, $\alpha = 0.025$

Model	Avg loss	Diff vs bench	Panel DM t	# better	# worse	Mean rank
HS	1.2439	0.2332	5.71	0	12	20.0
GARCH-N	1.0783	0.0660	5.38	0	12	18.3
GARCH-t	1.0518	0.0404	4.57	0	10	15.8
GJR-skt	1.0103	0.0000		0	0	11.7
FHS	1.0087	-0.0015	-1.20	1	0	10.1
GARCH-EVT	1.0082	-0.0019	-1.28	2	0	9.9
CAViaR-ES	1.0097	-0.0016	-0.30	0	1	10.3
FZ-1F	1.0806	0.0702	10.86	0	12	18.4
FZ-Hyb	1.0535	0.0433	7.54	0	10	16.2
LSTM-FZ	1.0032	-0.0081	-1.03	0	0	7.9
TRF-FZ	1.0380	0.0263	1.85	0	2	15.2
GJR-LSTM	0.9962	-0.0146	-2.76	5	0	5.2
NG-FZ	1.0035	-0.0076	-1.26	0	0	7.4
GJRX-skt	**0.9873**	-0.0231	-3.88	5	0	4.5
LSTMQ-FZ	1.0012	-0.0104	-1.38	0	0	7.0
CTG-FZ	1.0009	-0.0106	-1.43	0	0	6.2
COMB-Econ	1.0129	0.0025	0.98	0	0	12.0
COMB-All	0.9989	-0.0115	-3.99	6	0	6.0
COMB-All+NG	0.9969	-0.0135	-4.55	7	0	4.7
COMB-All+X	0.9935	-0.0169	-5.82	8	0	**3.2**

Avg loss: FZ0 loss averaged over days and then over the 12 assets. Diff vs bench: average loss minus that of GJR-skt (negative = better); Panel DM t: Diebold–Mariano statistic on the cross-asset average loss differential (Newey–West). # better / worse: assets where the single-asset DM test is significant at 5%. Mean rank across assets (1 = best).

univariate econometric model, even without the foreign news.

Table 4: Holdout sample: Model confidence set (90%) membership

Model	# assets 1%	# assets 2.5%	# assets 5%	Pooled p 1%	Pooled p 2.5%	Pooled p 5%
HS	0	0	0	0.000	0.000	0.000
GARCH-N	0	0	0	0.000	0.000	0.000
GARCH-t	2	1	1	0.001	0.000	0.000
GJR-skt	10	5	4	0.002	0.000	0.000
FHS	9	6	4	0.009	0.000	0.000
GARCH-EVT	9	6	4	0.009	0.000	0.000
CAViaR-ES	11	10	11	0.066	0.025	0.045
FZ-1F	0	0	0	0.000	0.000	0.000
FZ-Hyb	1	0	0	0.000	0.000	0.000
LSTM-FZ	11	12	12	0.278	0.508	0.540
TRF-FZ	9	8	8	0.038	0.025	0.029
GJR-LSTM	11	12	12	0.300	0.567	0.592
NG-FZ	10	12	11	0.101	0.252	0.540
GJRX-skt	12	12	12	1.000	1.000	1.000
LSTMQ-FZ	11	12	12	0.300	0.567	0.552
CTG-FZ	11	12	12	0.300	0.567	0.592
COMB-Econ	6	2	2	0.000	0.000	0.000
COMB-All	5	3	2	0.001	0.000	0.000
COMB-All+NG	10	9	7	0.024	0.008	0.003
COMB-All+X	11	12	12	0.300	0.567	0.592

Columns '# assets': number of assets whose MCS contains the model. 'Pooled p': MCS p-value on the cross-asset average loss series; a model is in the pooled MCS if $p > 0.1$. Range statistic, stationary bootstrap, 2000 replications.

Calibration (Table 5) mirrors the first paper. GARCH-EVT and the combinations are the best calibrated, with violation rates of 2.5–2.6% and few rejections. GJRX-skt has the same calibration as GJR-skt and passes the exceedance-residual ES test in every market. The neural models are again slightly under-conservative. In regulatory terms (Table 6), COMB-All+X spends the fewest days in the Basel yellow zone of any model, with red-zone frequency as low as GJR-skt and no more capital.

6.4 Crisis periods

Table 7 shows that the value of cross-market information is not confined to crises. GJRX-skt is more accurate than GJR-skt in every subperiod, and significantly so in the calm 2013–2019 period. COMB-All+X is significantly more accurate in the global financial crisis, the euro-area crisis, the 2022 inflation shock and the calm period, with t between -3.2 and -4.0. The exception is the COVID-19 crash, where no difference is significant because the window is so short.

6.5 Why do the networks not use the information?

The trained contagion network does react to foreign news. In the development sample, switching the news off changes its Nikkei VaR by 0.45 percentage points on average, and the change is correlated 0.34 with the size of the U.S. move. The problem is therefore not that the network ignores the input. It fails to turn the input into a better forecast than a model that imposes the

Table 5: Holdout sample: Backtests, $\alpha = 0.025$: number of assets (out of 12) rejecting at 5%

Model	Hit %	UC	CC	DQ	ER	CCT-u	CCT-c	Z2
HS	3.21	12	12	12	12	12	12	-0.409
GARCH-N	3.56	12	12	12	12	12	12	-0.631
GARCH-t	3.42	11	11	12	5	11	10	-0.417
GJR-skt	2.68	2	2	2	4	0	0	-0.102
FHS	2.60	0	1	1	0	0	0	-0.060
GARCH-EVT	2.53	0	1	2	1	0	0	-0.033
CAViaR-ES	2.58	0	2	2	2	1	0	-0.067
FZ-1F	2.73	3	2	4	3	2	7	-0.110
FZ-Hyb	2.70	1	3	3	2	2	1	-0.107
LSTM-FZ	2.83	3	3	4	3	3	2	-0.156
TRF-FZ	2.81	2	4	12	7	4	2	-0.184
GJR-LSTM	2.73	3	3	4	3	3	1	-0.113
NG-FZ	2.61	0	0	3	5	3	2	-0.085
GJRX-skt	2.74	2	2	2	0	2	1	-0.110
LSTMQ-FZ	2.89	5	5	5	3	4	4	-0.180
CTG-FZ	2.88	5	5	4	3	4	3	-0.175
COMB-Econ	2.69	1	2	2	3	0	0	-0.110
COMB-All	2.64	0	1	1	2	0	0	-0.081
COMB-All+NG	2.63	0	1	1	1	0	0	-0.075
COMB-All+X	2.62	1	1	2	1	0	0	-0.071

Hit %: average violation rate (nominal 2.5%). VaR tests: UC Kupiec, CC Christoffersen conditional coverage, DQ Engle–Manganelli. ES tests: ER McNeil–Frey exceedance residuals (one-sided bootstrap), CCT-u/CCT-c Nolde–Ziegel unconditional/conditional calibration. Z2: Acerbi–Szekely statistic averaged over assets (negative = ES too optimistic).

Table 6: Holdout sample: Regulatory view: capital and Basel traffic light

Model	Capital (avg -ES 2.5%)	Yellow zone % days	Red zone % days
HS	3.09	34.8	4.9
GARCH-N	2.46	53.5	4.1
GARCH-t	2.76	31.8	0.3
GJR-skt	2.91	18.6	0.0
FHS	2.95	15.1	0.3
GARCH-EVT	2.96	14.0	0.0
CAViaR-ES	2.89	17.0	0.1
FZ-1F	3.08	19.2	2.2
FZ-Hyb	3.01	16.4	0.4
LSTM-FZ	2.85	17.4	0.3
TRF-FZ	2.87	22.1	0.3
GJR-LSTM	2.87	16.8	0.3
NG-FZ	2.84	20.6	0.9
GJRX-skt	2.88	18.9	0.1
LSTMQ-FZ	2.83	18.2	0.5
CTG-FZ	2.84	17.8	0.5
COMB-Econ	2.88	16.3	0.0
COMB-All	2.87	14.2	0.0
COMB-All+NG	2.87	13.9	0.0
COMB-All+X	2.87	13.5	0.0

Capital: average forecast of $-\mathrm{ES}_{2.5\%}$ (FRTB measure), in % of position value, averaged over assets. Traffic light: share of days on which the trailing 250-day count of 99% VaR violations is 5–9 (yellow) or $\geq$10 (red).

Table 7: Holdout sample: Loss difference vs GJR-skt by subperiod, $\alpha = 0.025$

	GFC	Euro crisis	COVID	Inflation 2022	Calm 2013-2019
HS	0.333 (3.0)	0.239 (3.0)	2.010 (1.8)	0.174 (2.3)	0.136 (3.5)
GARCH-N	0.119 (3.1)	0.058 (2.1)	0.363 (1.4)	0.121 (2.5)	0.032 (2.0)
GARCH-t	0.074 (2.5)	0.038 (1.9)	0.240 (1.2)	0.085 (2.5)	0.016 (1.3)
FHS	-0.008 (-1.6)	0.001 (0.3)	-0.003 (-0.2)	-0.006 (-1.3)	0.002 (0.8)
GARCH-EVT	-0.013 (-2.2)	0.000 (0.1)	-0.003 (-0.2)	-0.004 (-0.7)	0.002 (0.8)
CAViaR-ES	-0.003 (-0.1)	-0.004 (-0.5)	0.208 (1.7)	-0.022 (-1.4)	0.000 (0.0)
FZ-1F	0.129 (4.9)	0.065 (4.1)	0.226 (1.7)	0.078 (2.6)	0.065 (6.7)
FZ-Hyb	0.047 (2.5)	0.033 (3.3)	0.234 (1.6)	0.063 (2.8)	0.044 (5.9)
LSTM-FZ	-0.016 (-0.5)	-0.015 (-0.9)	0.055 (0.2)	-0.008 (-0.4)	-0.014 (-1.6)
TRF-FZ	-0.006 (-0.2)	-0.008 (-0.7)	0.608 (1.2)	-0.014 (-0.5)	0.013 (0.9)
GJR-LSTM	0.003 (0.1)	-0.012 (-1.1)	-0.114 (-1.4)	-0.022 (-1.2)	-0.005 (-0.6)
NG-FZ	-0.010 (-0.5)	-0.008 (-0.7)	-0.025 (-0.1)	-0.010 (-0.7)	-0.002 (-0.2)
GJRX-skt	-0.020 (-1.3)	-0.016 (-1.5)	-0.105 (-1.3)	-0.024 (-1.8)	-0.017 (-2.8)
LSTMQ-FZ	-0.030 (-1.1)	-0.017 (-1.1)	0.033 (0.1)	-0.018 (-0.9)	-0.013 (-1.6)
CTG-FZ	-0.030 (-1.1)	-0.017 (-1.1)	0.028 (0.1)	-0.018 (-0.9)	-0.014 (-1.7)
COMB-Econ	-0.002 (-0.2)	-0.001 (-0.1)	0.071 (0.9)	0.004 (0.5)	0.003 (0.8)
COMB-All	-0.027 (-3.8)	-0.014 (-3.2)	0.013 (0.1)	-0.017 (-2.3)	-0.007 (-2.4)
COMB-All+NG	-0.029 (-3.7)	-0.016 (-3.2)	-0.004 (-0.0)	-0.018 (-2.4)	-0.009 (-2.9)
COMB-All+X	-0.033 (-4.0)	-0.019 (-3.5)	-0.023 (-0.3)	-0.023 (-3.2)	-0.012 (-3.9)

Average FZ0 loss difference (negative = better than the benchmark); panel Diebold–Mariano t-statistic in parentheses.

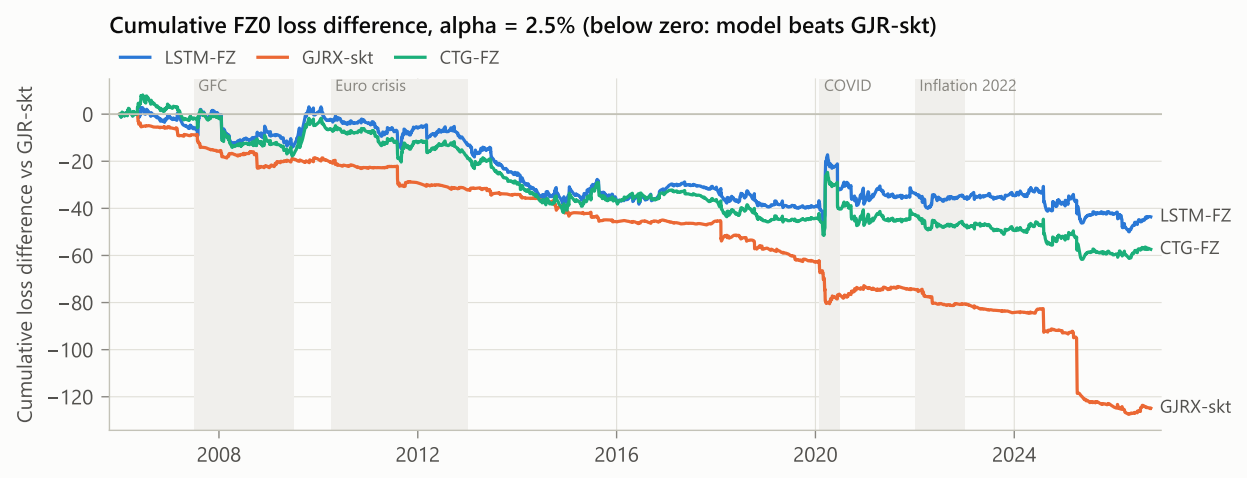


Figure 2: Holdout sample: cumulative FZ0 loss of LSTM-FZ, GJRX-skt and CTG-FZ minus GJR-skt (cross-asset average), $\alpha = 2.5\%$.

right functional form. Two pieces of evidence point to statistical efficiency. First, FZ0 learns from tail days only. In the development sample, adding the efficiency-improving QLIKE score halved the deficit of the market-specific neural GARCH (NG-FZ v2 to v3). Second, GJRX-skt embodies a strong and correct inductive bias: foreign squared news enters the variance additively, with two non-negative coefficients estimated by likelihood from every day. A network must instead discover this structure among roughly a dozen news inputs, market embeddings and attention weights, from the same eight-year windows. With daily data and tail-focused losses, the bias wins.

The learned news-impact curves of NG-FZ (Figure 3) are consistent with this reading. The network moves away from GJR-GARCH in an economically sensible direction: large positive shocks raise variance less than GJR implies. The correction is small, and it does not translate into significantly better forecasts.

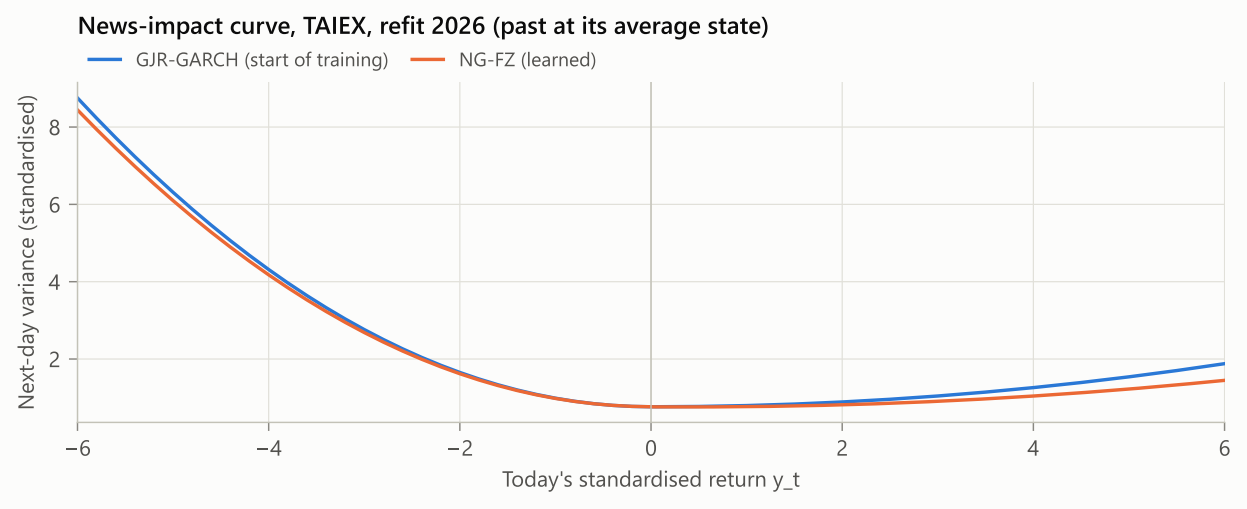


Figure 3: Holdout sample: news-impact curve of NG-FZ (refit January 2026, seed 0) for the Taiwan index versus the GJR-GARCH it is initialised at; the state is held at its training-sample average.

7 Conclusion

Giving tail-risk models the information of markets that have already closed improves forecasts substantially, and a two-parameter extension of GJR-GARCH captures most of the gain. The deep models we designed for the task did not. These were an attention-based contagion network, a neural GARCH with provable stability, and versions trained with an efficiency-improving composite score. They react to the news, but they do not turn it into better forecasts. The result holds in a development sample and, after a pre-registered freeze, in twelve markets that no model had seen. Combinations of econometric and neural forecasts remain the most accurate and the best calibrated, in line with our companion paper.

The mechanism, as far as our evidence identifies it, is statistical efficiency. FZ0 learns from tail days, while a likelihood learns volatility dynamics from every day. The correct functional form,

foreign squared news entering the variance additively, is easy to impose and hard to discover from eight-year windows. The broader lesson for machine learning in finance is that, in low signal-to-noise problems, the inductive bias and the information set deserve as much attention as the architecture. A deep model should be compared with the best structured model *given the same information*, not with a model that has less.

Limitations and extensions. We use daily close-to-close returns of equity indices and a one-day horizon. Intraday data would sharpen the timing of transmission, and other asset classes would test its generality. Larger pooled samples, for example thousands of individual stocks, may change the efficiency trade-off in favour of flexible models. The finding that tail risk flows mainly from the West to Asia overnight invites a structural follow-up: whether the transmitted component of tail risk is priced, in the spirit of Kelly and Jiang (2014) and Bali et al. (2009).

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A Additional tables

Table 8: Holdout sample: Average FZ0 loss, $\alpha = 0.01$

Model	Avg loss	Diff vs bench	Panel DM t	# better	# worse	Mean rank
HS	1.5709	0.3393	5.38	0	12	20.0
GARCH-N	1.3914	0.1565	5.53	0	12	18.7
GARCH-t	1.2861	0.0540	3.88	0	8	15.2
GJR-skt	1.2310	0.0000		0	0	11.2
FHS	1.2277	-0.0033	-1.34	1	0	9.2
GARCH-EVT	1.2257	-0.0053	-1.85	2	0	8.9
CAViaR-ES	1.2289	-0.0033	-0.42	0	1	8.6
FZ-1F	1.3837	0.1521	9.85	0	11	18.2
FZ-Hyb	1.3017	0.0703	6.01	0	7	16.1
LSTM-FZ	1.2261	-0.0063	-0.51	0	0	8.4
TRF-FZ	1.2789	0.0455	2.06	0	2	14.8
GJR-LSTM	1.2155	-0.0164	-1.92	2	0	6.2
NG-FZ	1.2289	-0.0029	-0.27	0	0	9.2
GJRX-skt	**1.1936**	-0.0375	-3.51	4	0	4.2
LSTMQ-FZ	1.2230	-0.0098	-0.82	1	0	7.7
CTG-FZ	1.2225	-0.0102	-0.85	1	0	7.0
COMB-Econ	1.2346	0.0034	0.70	0	1	11.7
COMB-All	1.2170	-0.0143	-2.78	3	0	6.5
COMB-All+NG	1.2143	-0.0169	-3.16	4	0	4.8
COMB-All+X	1.2088	-0.0225	-4.28	6	0	**3.4**

Avg loss: FZ0 loss averaged over days and then over the 12 assets. Diff vs bench: average loss minus that of GJR-skt (negative = better); Panel DM t: Diebold–Mariano statistic on the cross-asset average loss differential (Newey–West). # better / worse: assets where the single-asset DM test is significant at 5%. Mean rank across assets (1 = best).

Table 9: Holdout sample: Average FZ0 loss, $\alpha = 0.05$

Model	Avg loss	Diff vs bench	Panel DM t	# better	# worse	Mean rank
HS	0.9966	0.1726	5.89	0	12	20.0
GARCH-N	0.8598	0.0355	5.52	0	11	18.2
GARCH-t	0.8557	0.0315	4.98	0	10	17.1
GJR-skt	0.8233	0.0000		0	0	10.8
FHS	0.8225	-0.0007	-0.73	0	0	11.1
GARCH-EVT	0.8223	-0.0009	-0.95	0	0	11.0
CAViaR-ES	0.8212	-0.0028	-0.75	0	0	9.3
FZ-1F	0.8556	0.0318	8.06	0	10	16.8
FZ-Hyb	0.8478	0.0243	6.62	0	9	15.8
LSTM-FZ	0.8182	-0.0060	-0.92	0	0	8.3
TRF-FZ	0.8425	0.0183	1.67	0	4	15.6
GJR-LSTM	0.8126	-0.0109	-3.05	6	0	5.0
NG-FZ	0.8155	-0.0084	-2.04	0	0	7.1
GJRX-skt	**0.8072**	-0.0161	-4.00	6	0	4.3
LSTMQ-FZ	0.8154	-0.0089	-1.55	1	0	7.1
CTG-FZ	0.8151	-0.0092	-1.61	1	0	6.2
COMB-Econ	0.8250	0.0015	0.83	0	0	12.2
COMB-All	0.8142	-0.0093	-4.17	7	0	6.4
COMB-All+NG	0.8125	-0.0110	-4.89	7	0	4.4
COMB-All+X	0.8099	-0.0135	-6.27	8	0	**3.2**

Avg loss: FZ0 loss averaged over days and then over the 12 assets. Diff vs bench: average loss minus that of GJR-skt (negative = better); Panel DM t: Diebold–Mariano statistic on the cross-asset average loss differential (Newey–West). # better / worse: assets where the single-asset DM test is significant at 5%. Mean rank across assets (1 = best).

Table 10: Development sample: Pre-registered hypothesis tests (FZ0 loss of first model minus second)

Test (level)	Loss diff	Panel DM t	p-value	# assets better	# assets worse
H1: CTG-FZ vs GJRX-skt (1%)	0.0178	0.90	0.368	0	1
H1: CTG-FZ vs GJRX-skt (2.5%)	0.0153	1.22	0.221	0	2
H1: CTG-FZ vs GJRX-skt (5%)	0.0150	1.64	0.101	0	2
H2: CTG-FZ vs LSTM-FZ (1%)	0.0040	1.16	0.248	0	0
H2: CTG-FZ vs LSTM-FZ (2.5%)	-0.0007	-0.29	0.773	0	0
H2: CTG-FZ vs LSTM-FZ (5%)	-0.0058	-2.23	0.026	2	0
H3: COMB-All+X vs COMB-All (1%)	-0.0083	-4.39	0.000	10	0
H3: COMB-All+X vs COMB-All (2.5%)	-0.0059	-6.11	0.000	12	0
H3: COMB-All+X vs COMB-All (5%)	-0.0046	-7.18	0.000	12	0
Ablation: CTG-FZ vs LSTMQ-FZ (1%)	0.0001	0.09	0.927	0	3
Ablation: CTG-FZ vs LSTMQ-FZ (2.5%)	0.0002	0.48	0.632	0	1
Ablation: CTG-FZ vs LSTMQ-FZ (5%)	-0.0000	-0.05	0.961	0	0
Info: GJRX-skt vs GJR-skt (1%)	-0.0220	-2.93	0.003	3	0
Info: GJRX-skt vs GJR-skt (2.5%)	-0.0193	-3.90	0.000	5	0
Info: GJRX-skt vs GJR-skt (5%)	-0.0161	-4.60	0.000	7	0

Negative loss difference = the first model is more accurate. Panel Diebold–Mariano test on the cross-asset average loss differential.

Table 11: Development sample: Average FZ0 loss, $\alpha = 0.025$

Model	Avg loss	Diff vs bench	Panel DM t	# better	# worse	Mean rank
HS	1.3353	0.2101	5.07	0	12	20.0
GARCH-N	1.1910	0.0646	4.64	0	10	17.8
GARCH-t	1.1692	0.0432	4.00	0	10	15.8
GJR-skt	1.1262	0.0000		0	0	9.2
FHS	1.1245	-0.0017	-0.93	0	0	8.2
GARCH-EVT	1.1237	-0.0021	-1.19	0	0	7.7
CAViaR-ES	1.1331	0.0060	0.98	0	0	10.5
FZ-1F	1.1925	0.0651	6.39	0	12	18.4
FZ-Hyb	1.1623	0.0377	4.00	0	6	15.9
LSTM-FZ	1.1244	-0.0033	-0.29	1	0	7.7
TRF-FZ	1.1497	0.0227	1.30	0	1	14.8
GJR-LSTM	1.1300	0.0036	0.61	0	0	10.2
NG-FZ	1.1412	0.0146	1.65	0	2	13.4
GJRX-skt	**1.1076**	-0.0193	-3.90	5	0	3.9
LSTMQ-FZ	1.1237	-0.0042	-0.40	1	0	7.8
CTG-FZ	1.1238	-0.0040	-0.38	1	0	7.9
COMB-Econ	1.1267	0.0007	0.17	0	0	9.5
COMB-All	1.1160	-0.0102	-2.05	3	0	4.5
COMB-All+NG	1.1161	-0.0101	-2.02	3	0	4.5
COMB-All+X	1.1102	-0.0160	-3.49	6	0	**2.3**

Avg loss: FZ0 loss averaged over days and then over the 12 assets. Diff vs bench: average loss minus that of GJR-skt (negative = better); Panel DM t: Diebold–Mariano statistic on the cross-asset average loss differential (Newey–West). # better / worse: assets where the single-asset DM test is significant at 5%. Mean rank across assets (1 = best).